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About Me

Ph.D. graduate in Software Engineering with expertise in machine learning, deep learning, and feature engineering, particularly for imbalanced data classification. Developed algorithms for simultaneous feature extraction and weighting, improving performance on high-dimensional and complex datasets. Applied these methods to healthcare and biomedical data, including electronic health records and medical imaging. Experienced in teaching and supervising student projects, with ongoing research in neural networks, high-dimensional data, and optimization. Eager to contribute to interdisciplinary AI projects and collaborate with international teams on advanced applications of representation learning.

Education

- Ph.D. Computer Engineering, Department of Computer, Islamic Azad University of Technical Engineering, Mashhad, Iran 2018-2023.

Thesis: *"Simultaneous Feature Extraction and Weighting by Maximizing Inter-Class Distance and Minimizing Intra-Class Distance in Imbalanced Classification problems"*.
Research advisor: Dr. Hossein Moattar & Dr. Yahya Forghan
- M.Sc. Computer Engineering, Department of Computer, Islamic Azad University of Technical Engineering, Ferdows, Iran 2011-2015.

Thesis: *"Presenting a Combined Method to Detect Fraud in Financial Statements and Credit Cards"*.
Research advisor: Dr. Hossein Moattar
- B.Sc. Computer Engineering, Computer Engineering, Department of Computer, Islamic Azad University of Technical Engineering, Mashhad, Iran 2005-2009.

Research Interest

Machine learning Algorithms (Classification & Clustering)

Deep learning (CNN)

Imbalanced data classification & optimization

Honors And Awards

2018: Ranked 9th for Ph.D exam, Islamic Azad University of Technical Engineering, Mashhad, Iran.
Patented reality-based systems developed in the treatment of obsessive-compulsive fear of cockroaches.
Awarded best idea at the festival of the best idea of health.

Professional Experience

- ❖ Khorasan Higher Education Center University- Mashhad Branch, Mashhad, Iran
Director of Graduate Studies

June 2025 – Present

- ❖ Advisor, Computer Science Student Association, Khorasan Higher Education Center, Mashhad, Iran, 2025 – Present
- ❖ Khorasan Higher Education Center University- Mashhad Branch, Mashhad, Iran
Assistant Professor
Sept 2024 - Present
- ❖ Islamic Azad University- Mashhad Branch, Mashhad, Iran
Lecturer
Mar 2020 - Jul 2024
- ❖ University Instructor of Computer Engineering, Department of Computer, Kashmir Higher Education Center, Iran
Lecturer
Dec 2022 – Jul 2024
- ❖ Researcher at Sara Institute of Smart Technology and Health, Islamic Azad University Mashhad, Iran 2021 – 2022
(<http://vcr.mshdiau.ac.ir/mainsara2020.html>)
- ❖ Participated as a reviewer, ICCKE 2021 and ICCKE 2022.
- ❖ Participated as a reviewer, ICCKE 2023 and ICCKE 2024.
- ❖ Participated as a reviewer, Asian Journal of Research in Computer Science, May 2025.
- ❖ Participated as a reviewer, Journal of Data Science and Intelligent Systems (JDSIS), September 2025.
- ❖ Participated as a reviewer, Journal of Data Science and Intelligent Systems (JDSIS), September 2025.
- ❖ Participated as a reviewer, Journal of Data Science and Intelligent Systems (JDSIS), October 2025.
- ❖ Participated as a reviewer, Journal of Data Science and Intelligent Systems (JDSIS), November 2025.
- ❖ Participated as a reviewer, Journal of Data Science and Intelligent Systems (JDSIS), November 2025.
- ❖ Participated as a reviewer, Journal of Data Science and Intelligent Systems (JDSIS), November 2025.
- ❖ Participated as a reviewer, Journal of Data Science and Intelligent Systems (JDSIS), December 2025.
- ❖ Participated as a reviewer, Journal of Data Science and Intelligent Systems (JDSIS), February 2026.
- ❖ Software manager at Reflection software company, Mashhad, Iran. Modares 2015 – 2016.
- ❖ Front-end designer at a Company published by Astan Quds Razavi, Mashhad, Iran. 2013-2014.
(<https://behnashr.ir/>)

Academic Teaching Experience

- ❖ University Instructor of Computer Engineering, Department of Computer, Khorasan Higher Education Center, Iran:
 - Advanced Algorithms (Master's level course)
 - Computer lab
 - Computer Fundamentals and Programming
 - Advanced Programming
 - Web Design
 - Advanced Programming Workshop
 - Project Work
 - Internship
 - Bachelor's Thesis Defense Judge
- ❖ University Instructor of Computer Engineering, Department of Computer, Kashmir Higher Education Center, Iran:
 - Advanced Computer Programming
 - Data Structures
 - Programming Languages
 - Logic Circuits
 - Analysis and Design of Algorithms
 - Theory of Languages and Automata
 - Web-Based Programming
 - Advanced Programming
 - Advanced Programming Workshop
 - Bachelor's Thesis Defense Judge
- ❖ Islamic Azad University- Mashhad Branch, Mashhad, Iran:
 - Computer lab
 - Computer Fundamentals and Programming

Technical and Language Skills

Language:

- English: Intermediate (B1–B2), Strong reading and writing; basic speaking skills. Currently attending English conversation classes to improve fluency.
- German: Beginner level (A1); currently learning and working to improve basic communication skills.
- Arabic: Basic knowledge
- Persian: Native speaker

Computer Skills:

- Programming & Data Science: Python (Pandas, NumPy, scikit-learn, Matplotlib), Matlab (Fuzzy, ANFIS, Neural Networks, Simulink, M-File)
- Machine Learning & AI: Knowledge of AI concepts, experience with ML techniques and optimization, familiarity with deep learning methods and libraries
- Other Tools: Git, SQL, Turtle, Kivy, Tkinter
- Web & Additional Skills: HTML, CSS, JavaScript, Django, WordPress

Publications (Scholar page)

1. **M. Fattahi**, M.H. Moattar. Y. Forghani, "*Locally alignment based manifold learning for simultaneous feature selection and extraction in classification problems*", **Knowledge-Based Systems** 2023, [https://doi.org/10.1016/j.knosys.2022.110088].
2. **M. Fattahi**, M.H. Moattar. Y. Forghani, "*Improved cost-sensitive representation of data for solving the imbalanced big data classification problem*", Journal of **Big Data** 2022, [https://doi.org/10.1186/s40537-022-00617-z].
3. **M. Fattahi**, M.H. Moattar. "*A New Hybrid Fraud Detection Approach in Credit Cards and Financial Statements*", **ICEEE** 2015, [https://www.researchgate.net/publication/390091965_A_New_Hybrid_Fraud_Detection_Approach_in_Credit_Cards_and_Financial_Statements].
4. **Book:** Fraud detection in financial statements and credit cards, book (March 2017) (Persian)
5. **Book:** Data analysis models and algorithms for intelligent data analysis, book (March 2018)
6. **Book:** JavaScript Programming for Beginners and Youth, book (March 2025) (Persian)
7. **Book:** Artificial Intelligence and Machine Learning for Coders, book (August 2025) (Persian)

References

- Associate Professor Mohamad. Hossein Moattar, Department of Computer, Islamic Azad University of Technical Engineering, Mashhad, Iran.
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Phone: +98 915 503 0540
- Associate Professor Yahya Forghani, Department of Computer, Islamic Azad University of Technical Engineering, Mashhad, Iran.
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